

Cryptography

Lecture 1

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August 24, 2026

Outline

- 1 Principles of Modern Cryptography (Ch. 1.1, 1.4)
- 2 Private-Key Encryption (Ch. 1.2)
- 3 Probability Review (Ch. A.3)
- 4 Perfectly-Secure Encryption (Ch. 2.1)
- 5 The One-Time Pad (Ch. 2.2)

A (Very) Brief History of Cryptography

- For 1000s of years, cryptography was “the art of writing or solving codes”
- Largely heuristic approaches to design codes, leading to a break-fix cycle
- Beginning in the 1970s, modern cryptography turned cryptography into a science, giving it a strong mathematical basis.



Principles of Modern Cryptography

- Formal definitions
- Precise assumptions
- Proofs of security

Kerckhoffs' Principle

“The cipher method must not be required to be secret, and it must be able to fall into the hands of the enemy without inconvenience.”

The Rule of Modern Crypto

No security by obscurity!

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- 5 The One-Time Pad (Ch. 2.2)

Private-key encryption

Alice



Bob



Private-key encryption



Key k



Key k

Private-key encryption

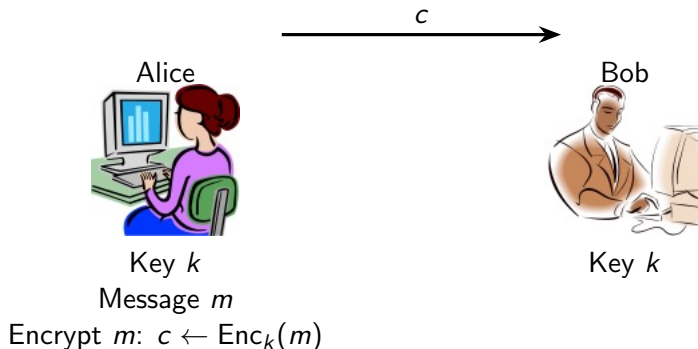


Key k
Message m

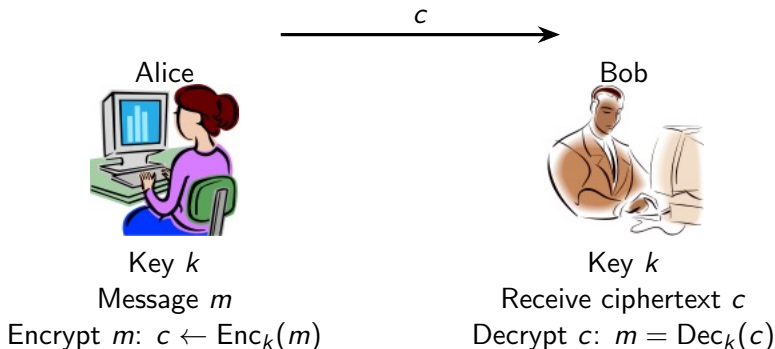


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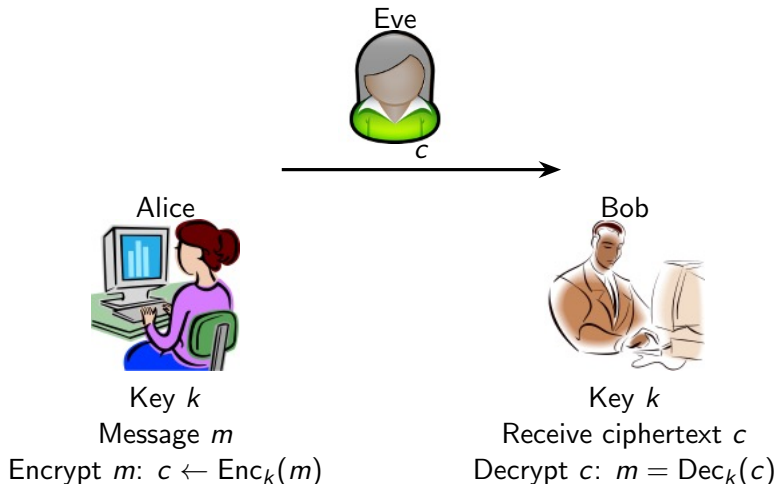
Private-key encryption



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Private-key encryption



Security – Informal

Eve gets to observe c , but can not learn m

Defining Encryption Functionality

Private-key (symmetric-key) encryption scheme:

- Gen: Outputs randomly chosen key k
- $\text{Enc}(k, m) : c \leftarrow \text{Enc}_k(m)$
- $\text{Dec}(k, c) : m = \text{Dec}_k(c)$

Terminology

m - plaintext

c - ciphertext

Correctness

For all k output by Gen and all messages m , $\text{Dec}_k(\text{Enc}_k(m)) = m$

Defining Encryption Security

Security Guarantee

What is a successful attack?

Threat Model

What can the adversary do?

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Regardless of what $\mathcal{A}$ knows about m , she learns no new information

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Time for a quiz and then a short break

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Definition

E_1 and E_2 are *independent* if $\Pr[E_1 \wedge E_2] = \Pr[E_1] \cdot \Pr[E_2]$

Bayes' Theorem

- Conditional Probability of E_1 given E_2 ($\Pr[E_2] > 0$):

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- Proof:

By definition of conditional probability,

$$\Pr[E_1 | E_2] \cdot \Pr[E_2] = \Pr[E_1 \wedge E_2] = \Pr[E_2 | E_1] \cdot \Pr[E_1].$$

$$\text{So, } \Pr[E_1 | E_2] = \frac{\Pr[E_2 | E_1] \cdot \Pr[E_1]}{\Pr[E_2]}$$

Law of Total Probability

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$$\Pr[A] = \sum_{i=1}^n \Pr[A \wedge E_i] = \sum_{i=1}^n \Pr[A \mid E_i] \cdot \Pr[E_i]$$

- Proof Sketch:

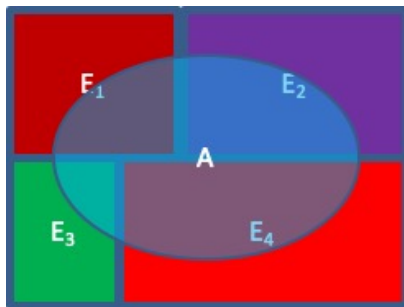


Some Useful Definitions and Facts

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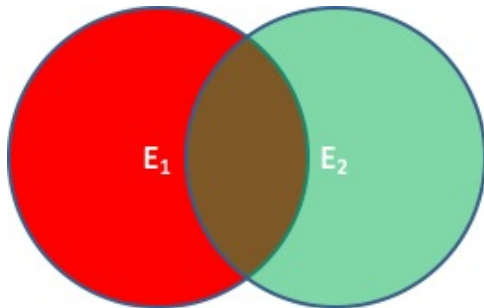
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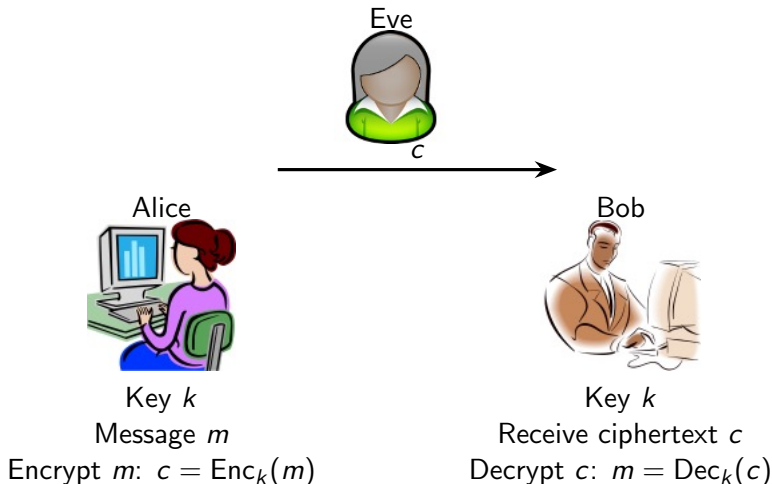
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Formalizing the Definition

Probability Distributions:

- Let M be a random variable denoting value of the message
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Remember

$\mathcal{M}$ is a space, M is a random variable, m is a value taken on by M
We will often look at $\Pr[M = m]$

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Def. 2 For all pairs $m, m' \in \mathcal{M}$ s.t. $|m| = |m'|$, for all $c \in \mathcal{C}$

$$\Pr[\text{Enc}_K(m) = c] = \Pr[\text{Enc}_K(m') = c]$$

A Game-Based Definition

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- The challenger chooses $k \leftarrow \text{Gen}$, $b \leftarrow \{0, 1\}$, computes $c \leftarrow \text{Enc}_k(m_b)$ and gives c to $\mathcal{A}$
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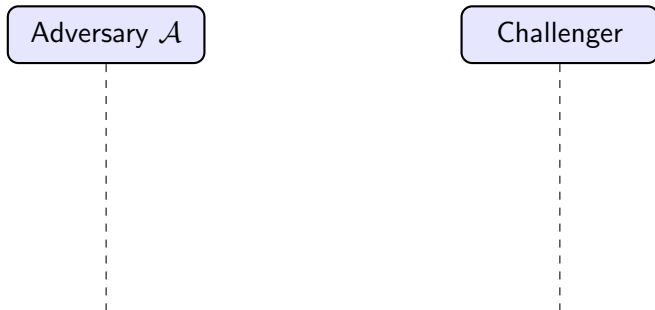
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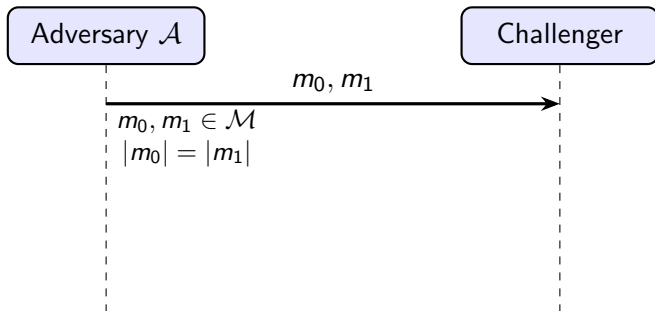
Definition: An encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ with message space $\mathcal{M}$ is *perfectly indistinguishable* if for all $\mathcal{A}$ it holds that

$$\Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}} = 1] = 1/2$$

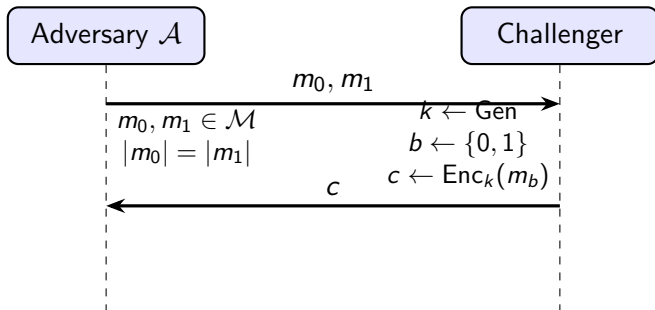
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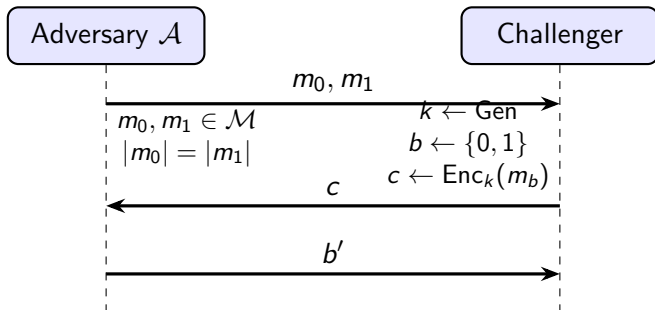
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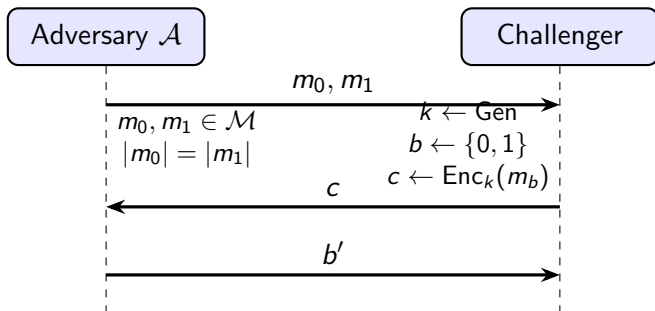
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Winning Condition

$b' = b$ Perfect indistinguishability: $\Pr[b' = b] = \frac{1}{2}$

We have seen three equivalent definitions of perfect secrecy:

- Message distribution unchanged after observing the ciphertext
- Identical ciphertext distributions for any two equal-length messages
- No adversary distinguishes their encryptions better than random guessing

Important

All three definitions are equivalent, and we will use them interchangeably.

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Let's Recall Our Goal – Perfect Secrecy

Informal Security Goal

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Encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ with message space $\mathcal{M}$ is *perfectly secret* if

- For all distributions over $\mathcal{M}$, for all $m \in \mathcal{M}$, for all $c \in \mathcal{C}$ with $\Pr[C = c] > 0$

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Correctness: For all $k \in \mathcal{K}$ and all $m \in \mathcal{M}$,

$$\text{Dec}_k(\text{Enc}_k(m)) = k \oplus (k \oplus m) = (k \oplus k) \oplus m = 0^\ell \oplus m = m$$

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Theorem: The OTP is perfectly secret ($\Pr[M = m \mid C = c] = \Pr[M = m]$)

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$$\begin{aligned}\Pr[C = c \mid M = m] &= \Pr[\text{Enc}_K(m) = c] = \Pr[m \oplus K = c] \\ &= \Pr[K = m \oplus c] = 2^{-\ell}\end{aligned}$$

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